University for the Creative Arts

BERLIN SCHOOL OF BUSINESS & INNOVATION

Essay / Assignment Title: TIME-SERIES FORECASTING OF HOUSEHOLD ENERGY CONSUMPTION USING MACHINE LEARNING MODELS

Programme title: Machine Learning Fundamentals

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CONTENTS

1. Introduction (LO1, LO2)
6

1.1 Background and Context
6

1.2 Limitations of Traditional Forecasting Methods
6

1.3 Role of Machine Learning in Time-Series Forecasting
6

1.4 Aim and Objectives of the Study
7

2. Literature Review (LO2)
8

2.1 Classical Time-Series Forecasting Approaches
8

2.2 Machine Learning Approaches for Energy Forecasting
8

2.3 Deep Learning Approaches (Brief Overview)
8

2.4 Research Gaps
9

3. Data Collection (LO1)
10

3.1 Dataset Description
10

3.2 Justification for Dataset Selection
10

4. Data Preprocessing (LO1)
11

4.1 Handling Missing Values and Data Cleaning
11

4.2 Date-Time Parsing and Indexing
11

4.3 Resampling Strategy
11

4.4 Feature Engineering
11

5. Exploratory Data Analysis (EDA) (LO1)
13

5.1 Descriptive Statistics
13

5.2 Trend Analysis
14

5.3 Seasonality and Pattern Analysis
16

6. Model Selection and Design (LO1, LO2)
21

6.1 Train–Test Split Strategy
21

6.2 Selected Machine Learning Models
21

6.3 Justification of Model Choices
21

7. Code Implementation (LO1)
22

7.1 Implementation Environment
22

7.2 Model Implementation Workflow
22

8. Model Evaluation and Results (LO1, LO2)
23

8.1 Evaluation Metrics
23

8.2 Model Performance Comparison
23

8.3 Visual Evaluation
25

8.4 Best-Performing Model
28

9. Conclusion and Future Work (LO2)
31

9.1 Summary of Findings
31

9.2 Practical Implications
31

9.3 Limitations
31

9.4 Future Improvements
32

Reference List
32

Appendices
34

Appendix 1:
34

Google Colab Link
34

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1. Introduction (LO1, LO2)

1.1 Background and Context

Energy consumption forecasting is an important feature of contemporary energy management systems, and it requires precise household energy consumption forecasting. Predictions of demand with high accuracy help in efficient allocation of resources as well as minimise operational costs and increase supply reliability. Forecasting in the smart grid environment makes it possible to implement dynamic demand response and load balancing strategies. Moreover, precise forecasts support sustainability goals by reducing the amount of energy wasted and assist in introducing sources of renewable energy. According to Kim *et al.* (2023), such forecasting of consumption at the household level enhances energy planning and superior decision-making in the energy systems.

1.2 Limitations of Traditional Forecasting Methods

Stress on traditional forecasting methods makes huge assumptions on linear assumptions and predefined statistical structures. These methods are frequently not effective in capturing complicated consumption patterns that are non-linear and exhibit a sudden switch in demand. There are also complications with seasonal patterns at various levels of time, and this complicates model accuracy. Also, the classical methods have scalability problems when dealing with high-frequency and large datasets. According to Ghazal (2022), a pure statistical model can often be unproductive in a state of the art energy system since they lack the versatility to adapt to the changing and diverse consumption profiles in the residential energy forecasting environment of the current era.

1.3 Role of Machine Learning in Time-Series Forecasting

Machine learning models provide adaptable data-driven structures with the power to model multifaceted temporal transformations in the energy consumption data. The models are non-linear learners who are not constrained by distributional assumptions based on historical

observations. Moreover, machine learning algorithms have the capability of combining numerous information-explaining features with the aim of improving predictability. Olu-Ajayi *et al.* (2022) have postulated that forecasting machine learning models are superior to traditional models when dealing with residential energy data with high dimensions and high temporal correlation. Such ability renders them applicable to intricate household electricity consumption forecasting activities in actuality.

1.4 Aim and Objectives of the Study

Aim:

This research aims to create and assess machine learning models to predict household energy consumption tasks very well.

Objectives:

- To pre-process and analyse the household electricity time-series data to determine the meaningful patterns and trends of temporal consumption.
- To compare various machine learning models on short-term household power demand prediction based on past data.
- To assess the performance of the models with standard error measures and analyse the forecasting performance critically with an aim of making decisions.

2. Literature Review (LO2)

2.1 Classical Time-Series Forecasting Approaches

Hossain *et al.* (2024) reviewed the problem of time-series analytics (along with machine learning models) and their application to smart grid energy demand forecasting. The analysis involved historical consumption trends in order to enhance a better short-term load prediction. Their results showed that time structure is a key factor in determining the reliability of residential energy systems in the world. This work educates the ongoing study by notifying the underlying principles in time series models, and revealing the shortcomings in the conventional structure to respond to non-linear consumption dynamics in households in the context of realistic forecasting in energy systems.

2.2 Machine Learning Approaches for Energy Forecasting

In Ghazal (2022), the author examined fused machine learning in energy demand forecasting by integrating several predictive models. The research showed that forecasting ability was much higher than single-model methods. The outcome was that hybrid regression-based and ensemble methods were useful in encapsulating multi-faceted consumption behaviours in giant energy datasets of a variety of residential and grid uses. The present study contributes to the research in terms of keeping the momentum of machine learning models that clearly demonstrate their value when dealing with non-linearity and scale in household energy forecasting tasks in the modern data-driven energy systems frameworks.

2.3 Deep Learning Approaches (Brief Overview)

Olu-Ajayi *et al.* (2022) experimented with the idea of deep learning, such as recurrent neural networks, in the prediction of residential building energy consumption. The experiment showed that the LSTM and its variants successfully represent the temporal dependence over a long period of time. The experimental evidence showed that it could predict better than traditional machine learning algorithms, though with more computing power in large-scale energy

forecasting. This paper is informative as it justifies the deep learning relevance and strengthens the argument to use a balanced model complexity in real-life household energy forecasting environments within the conditions of real-world data availability and resource constraints.

2.4 Research Gaps

Although they have made tremendous improvements, the current research is mostly limited to one model performance or a short time range, which limits the scope of generalisability. Most of the methods put emphasis on accuracy, as opposed to interpretability and practical deployment concerns. Moreover, the approach of feature engineering is not uniformly utilised in the literature. According to Wen and Liu (2025), systematic feature selection is an unexplored area of residential energy forecasting-based research. Therefore, the current project employs machine learning models that have preprocessing structures to deal with the gaps of scalability, interpretability, and real-world applicability to modern realities of smart grids and home energy management in the world today.

3. Data Collection (LO1)

3.1 Dataset Description

The data provided in this research was found in Kaggle and belongs to the UCI Machine Learning Repository. It contains records of household electricity consumption at intervals of one minute distributions over 4 years, containing over two million observations. The major variables represented will be global active power, reactive power, voltage, level of current flow and sub-metered energy consumption within the limits of the household. According to Kim *et al.* (2023), the high-resolution smart meter data categories allow studying more closely the dynamics of the domestic consumption process. This data is thus a good empirical base for prediction analysis.mn

3.2 Justification for Dataset Selection

This data is commonly considered a gold standard when it comes to energy forecasting studies because of the size of the dataset, the time scale and real-world applicability. The duration of the time series permits trend and seasonality analysis, whereas measurements at the minute level in the time series permit aggregation in a flexible way over time. According to Ghazal (2022), benchmark energy datasets have the advantage of improving comparative methodological reproducibility in machine learning research. As a result, the dataset can be effectively used in the evaluation of time-series forecasting models as well as the analysis of their performance in practise in households regarding household energy consumption

4. Data Preprocessing (LO1)

4.1 Handling Missing Values and Data Cleaning

There are missing values in the dataset due to the inability to connect to the sensor and an interruption in recording the data. The strategy of structured data cleaning was used with removing records that are incomplete at the first stage, and forward and backwards filling after resampling of time. This method maintained continuity, hence reducing the loss of information. According to Wen and Liu (2025), whenever there is temporal integrity in energy time-series modelling, it is crucial to have consistent missing value management. Thus, the chosen approach ensured the reliability of data, and sound training and evaluation of the machine learning model downstream were facilitated.

4.2 Date-Time Parsing and Indexing

Date and time variables were merged together, and a single variable of timestamp was created, which was afterwards attributed as an index of the dataset. To maintain the chronological order and complete rolling analysis and resampling, it is critically important to create a time-based index in place of the chronological order. According to Ali *et al.* (2024), proper alignment of the timestamps enhancesthe understanding and accuracy of forecasting on smart grid analytics. As a result, valid time-series retention and chronological training -testing data partitioning were made possible by explicitly retaining the temporal dependencies via this preprocessing step.

4.3 Resampling Strategy

The initially minute-level data have been upsampled to a frequency of one hour in order to cut down noise and computational costs. According to Hossain *et al.* (2024), temporal aggregation has the benefit of enhancing the stability of forecasting but without disrupting the key demand trends, making the models more efficient and interpretable.

4.4 Feature Engineering

The timestamp was converted to time-based features, such as hour, day of week, and month. According to Wilfling (2023), the temporal feature engineering leads to improved learning of models by revealing cyclic consumption behaviour within the data of household consumption of energy.

5. Exploratory Data Analysis (EDA) (LO1)

5.1 Descriptive Statistics

The level of active power and voltage around the world showed great variation as observed in the descriptive statistics, and this indicates various household usage habits. Central tendency measures showed moderate levels of mean consumption, whereas dispersion values identified periods of infrequent demand. According to Kamoona *et al.* (2023), statistical summaries set the essential background for the interpretation of the variability in itself in the context of energy consumption before the modelling of predictive forecasting. The analysis provided a numerical point of reference that explains consumption distributions over the observation period.

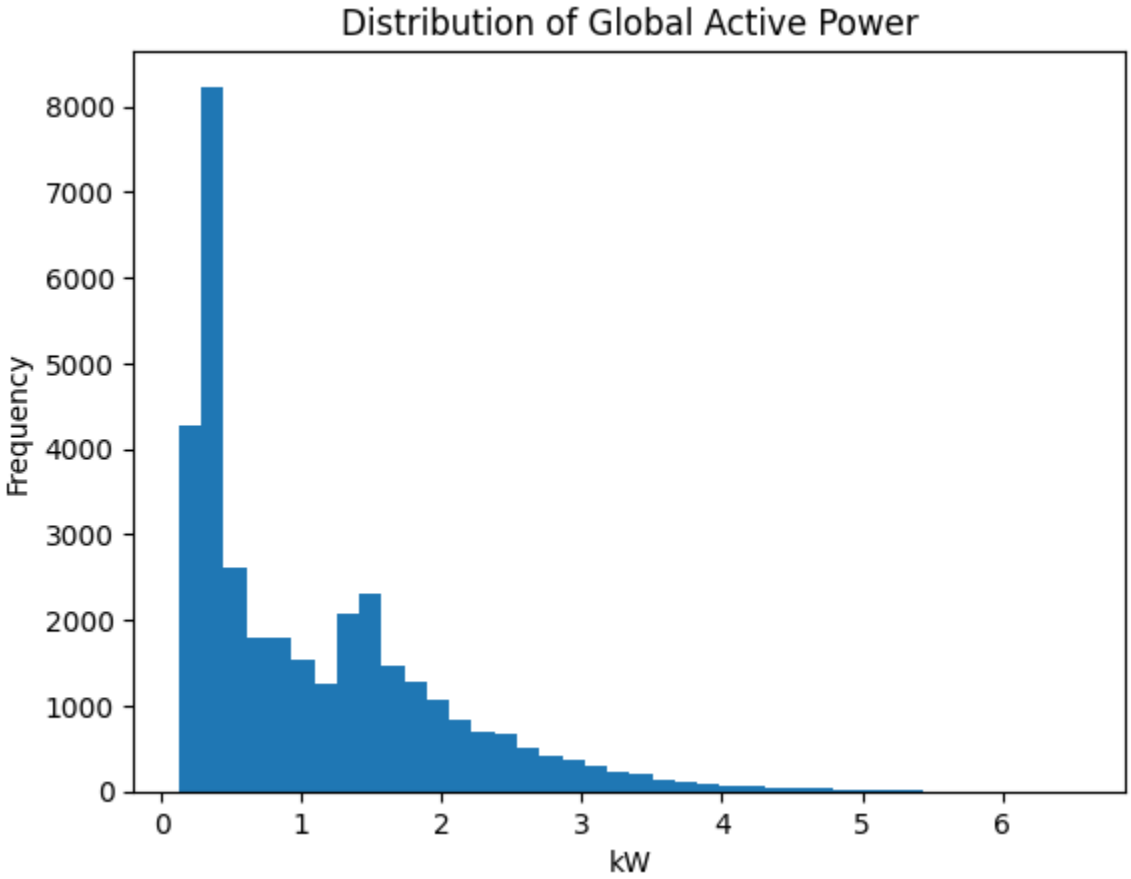


Figure 1: Distribution of Global Active Power

(Source: Self-created in Google Colab)

Figure 1 shows the frequency distribution of the active power consumption of hours on Earth. The mean for the consumption (0.805 kW) means that the consumption is smaller than the mean, which is a right-skewed distribution. This brings out the occasional high-demand and justifies the need for models that are resilient to non-normal data.

5.2 Trend Analysis

Trend analysis showed that there were relatively constant fluctuations in household power consumption over the long term, and relatively constant voltage behaviour. Periodic levels of active power were raised throughout the world, associated with the increase in household activity. According to Alzoubi (2022), long-term trend analysis helps determine the pattern of stability of the consumption and the risk of an increase in demand in the smart home context. This analysis supported the existence of underlying historical patterns needed in the development of successful forecasting models.

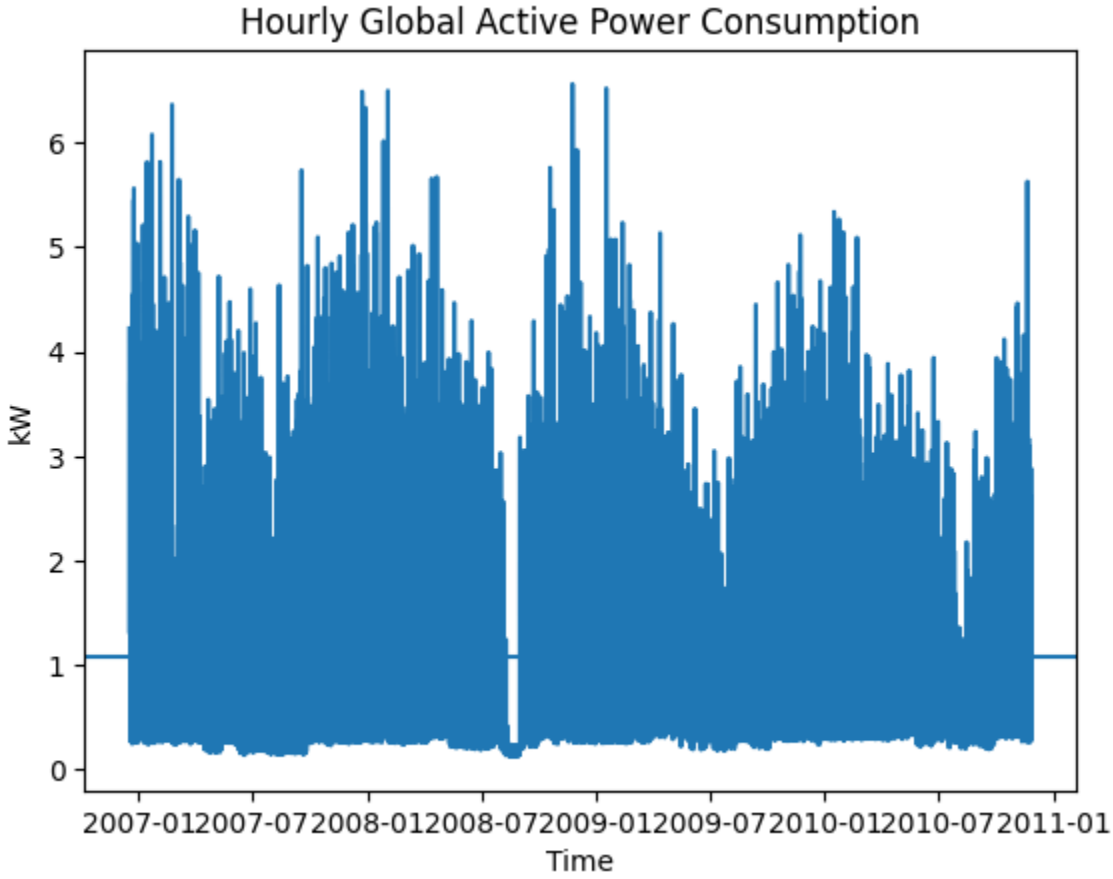


Figure 2: Hourly Global Active Power Consumption

(Source: Self-created in Google Colab)

This value is used to demonstrate the amount of electricity used in households on a day-in, day-out basis throughout the entire observation time. The mean active power all over the world is 1.089 kW, as shown by the horizontal reference line. The time-series stands considerably and deterministically variable, showing how households have changed in usage patterns therefor,e time-series modelling would be inspirational in the precise forecasting of energy demand.

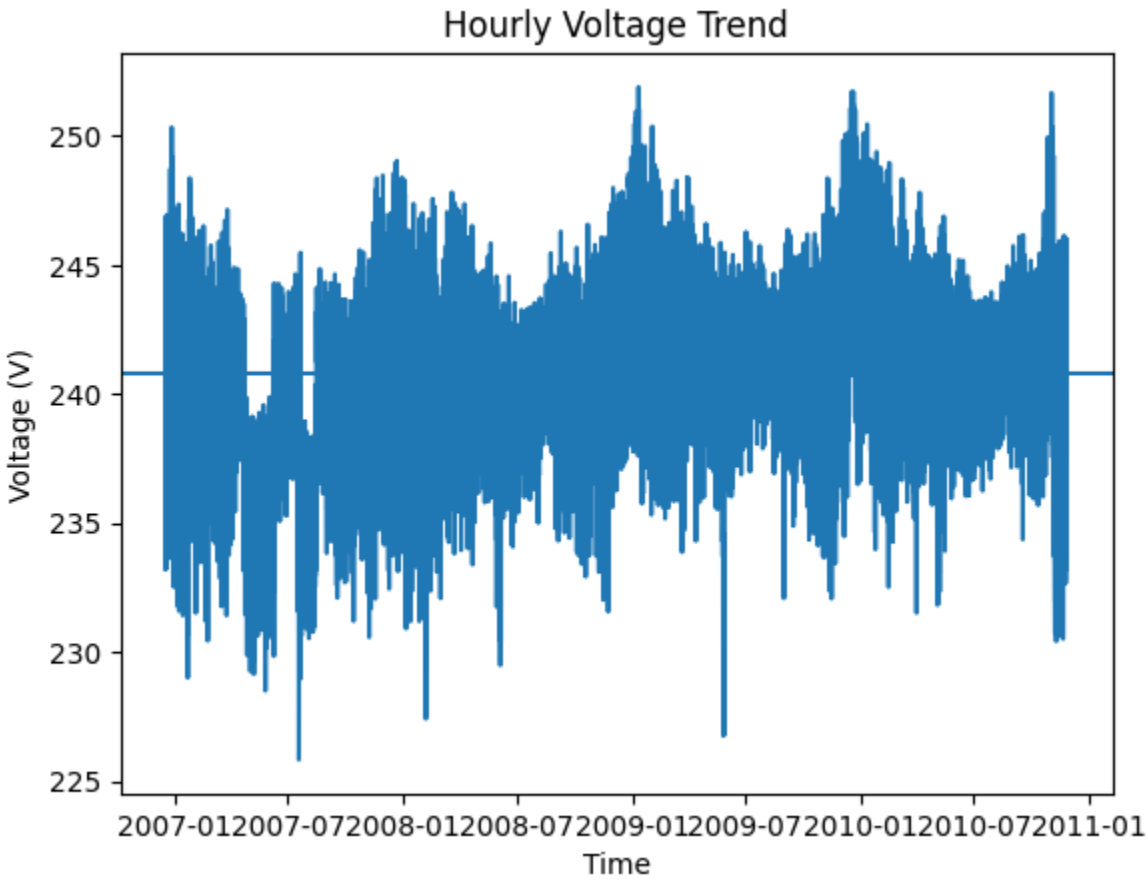


Figure 3: Hourly Voltage Trend

(Source: Self-created in Google Colab)

The voltage levels per hour were captured and displayed in Figure 3. The average electric potential of 240.84 V indicates the survival of the overall stability of the system and a slight change with time. Stable voltage behaviour not only helps to prove that the dataset is reliable, but

also means that any changes in power consumption are caused by changes in demand and are not caused by the supply.

5.3 Seasonality and Pattern Analysis

The visual analysis revealed repeated patterns of daily, weekly and monthly consumption, which meant that there was high temporal seasonality. The usage was highest at certain hours and at week timelines. According to Olu-Ajayi *et al.* (2022), the awareness of the seasonal consumption cycles leads to a higher predictive power in residential energy prediction. Therefore, these lessons explained the temporal features inclusion and further machine learning model choice/ evaluation strategies.

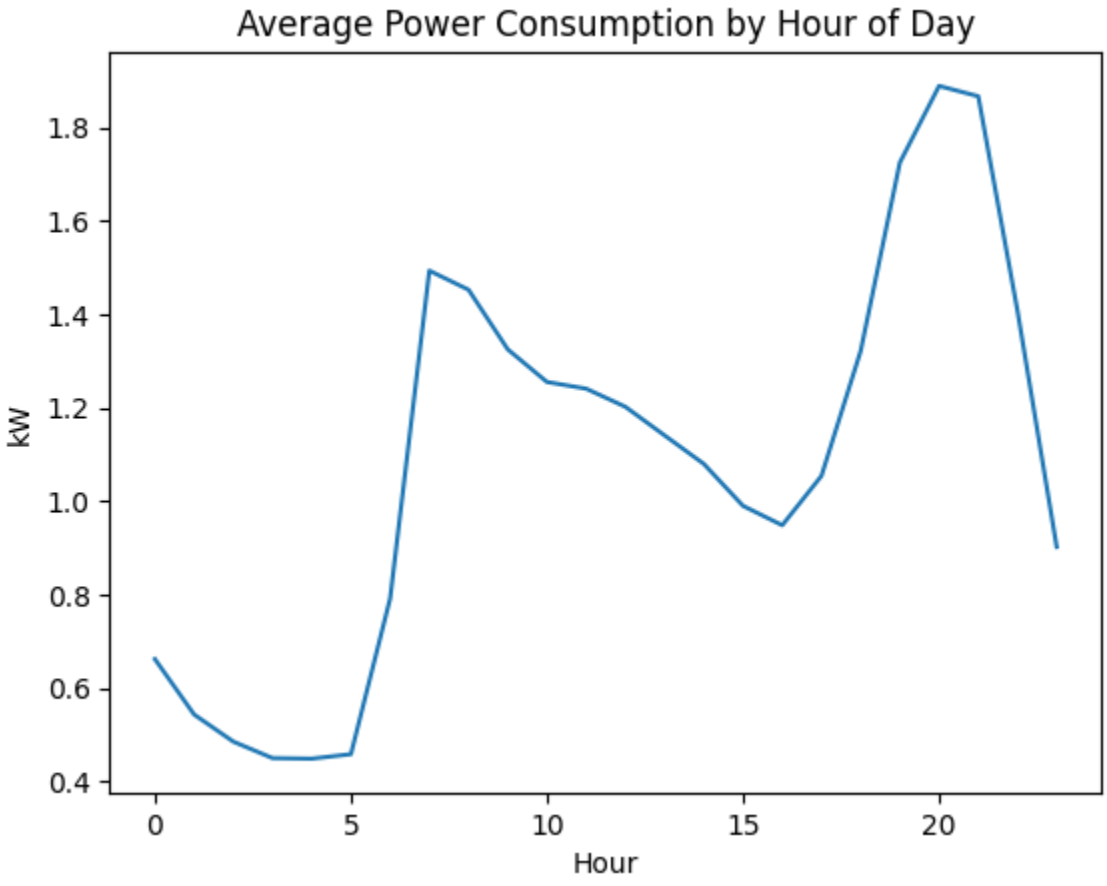


Figure 4: Average Power Consumption by Hour of Day

(Source: Self-created in Google Colab)

This number emphasises the diurnal consumption trends through averaging the global active power hour by hour. The peak demand is set to reach 1.889 kW, which means that it is more used during particular periods of the day. These apparent hourly trends justify the consideration of time-based characteristics and portray high intraday seasonality in the energy consumption of households.

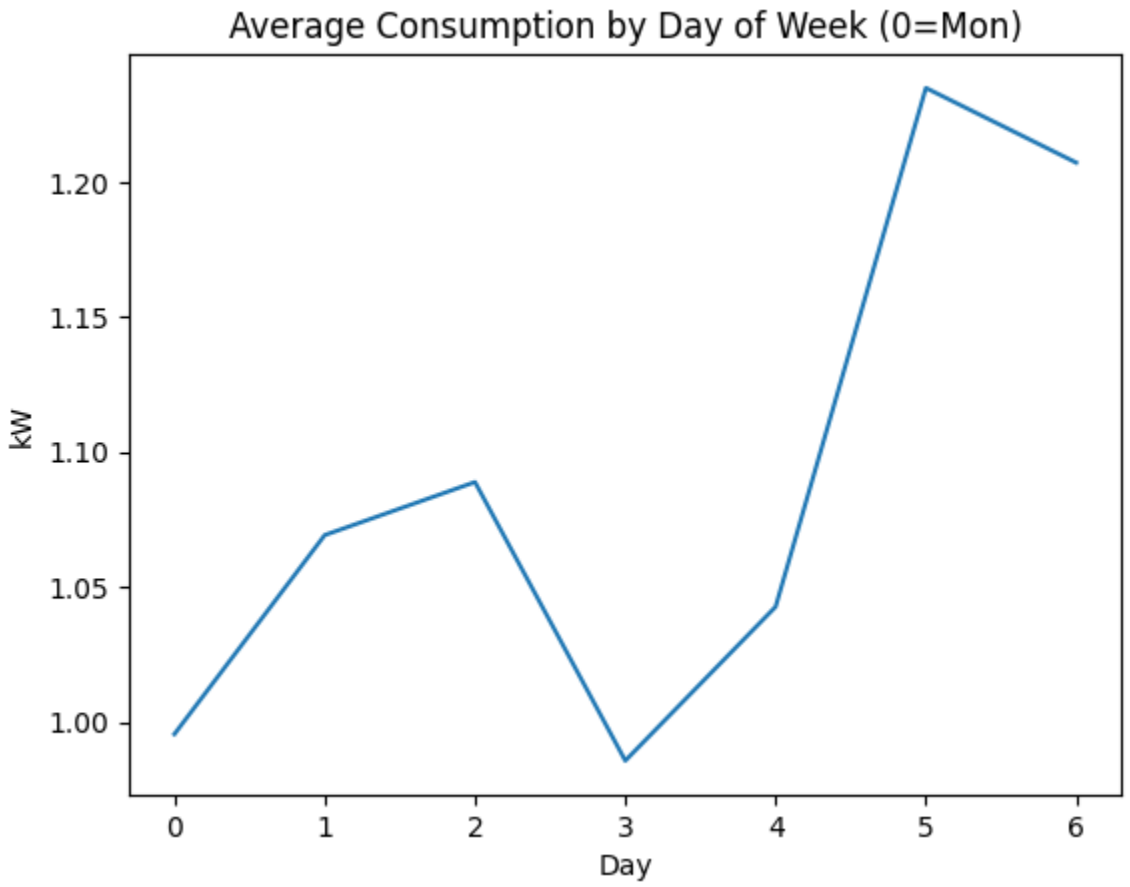


Figure 5: Average Consumption by Day of Week (0+ Mon)

(Source: Self-created in Google Colab)

Figure 5 is used to compare average energy usage during weekdays. The maximum consumption on weekdays is 1.235 kW, and this shows that there is a moderate day-to-day variation. Such trends indicate behavioural factors on energy demand and explain the consideration in the predictive modelling framework of the day-of-week aspect.

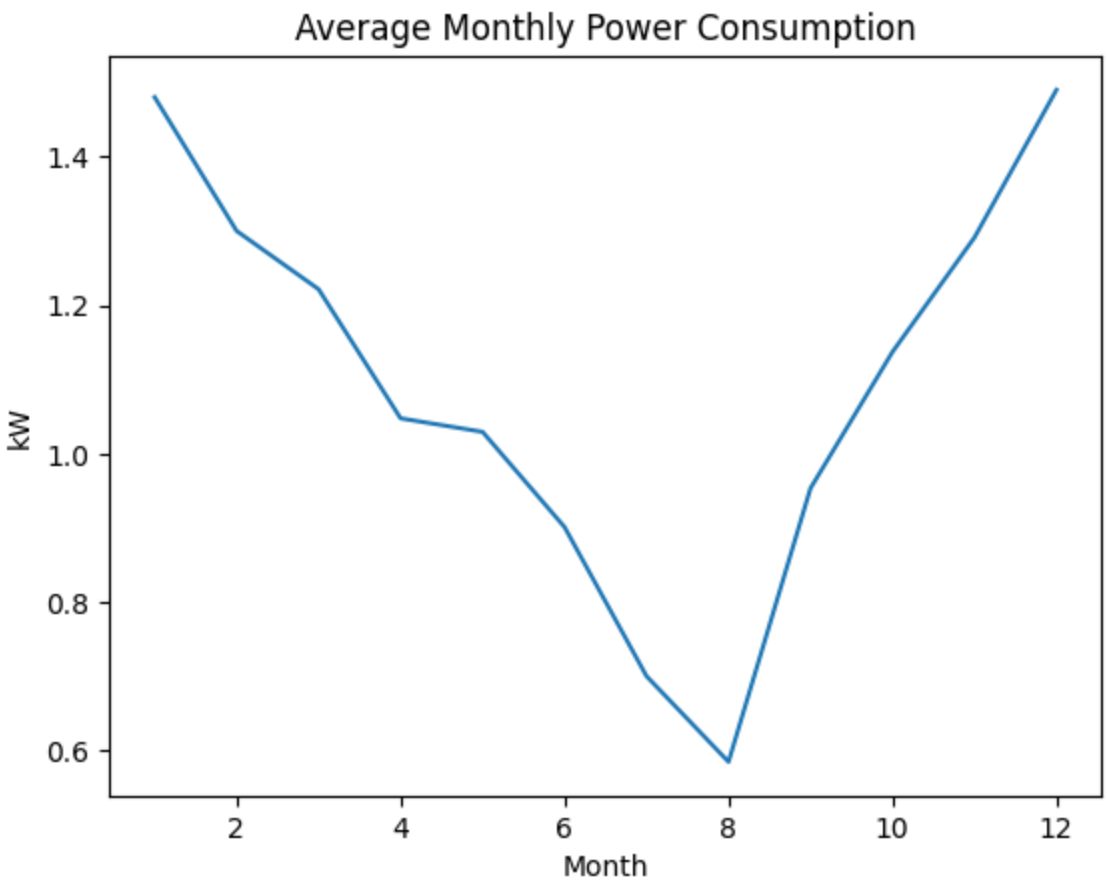


Figure 6: Average Monthly Power Consumption

(Source: Self-created in Google Colab)

Figure 6 shows the difference in the seasonal electricity consumption (monthly). The maximum average monthly consumption is 1.49 kW, which is evidence of the seasonal demand, which may be caused by the need to heat or cool. This validates the monthly features in models of energy forecasting that have long-term dynamics of the season.

```
# Model 3 – Support Vector Regression (SVR)


# SECTION 6 (ADDITION): MODEL 3 – SUPPORT VECTOR REGRESSION

from sklearn.svm import SVR
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

svr_model = Pipeline([
    ("scaler", StandardScaler()),
    ("svr", SVR(kernel="rbf", C=10, epsilon=0.1))
])

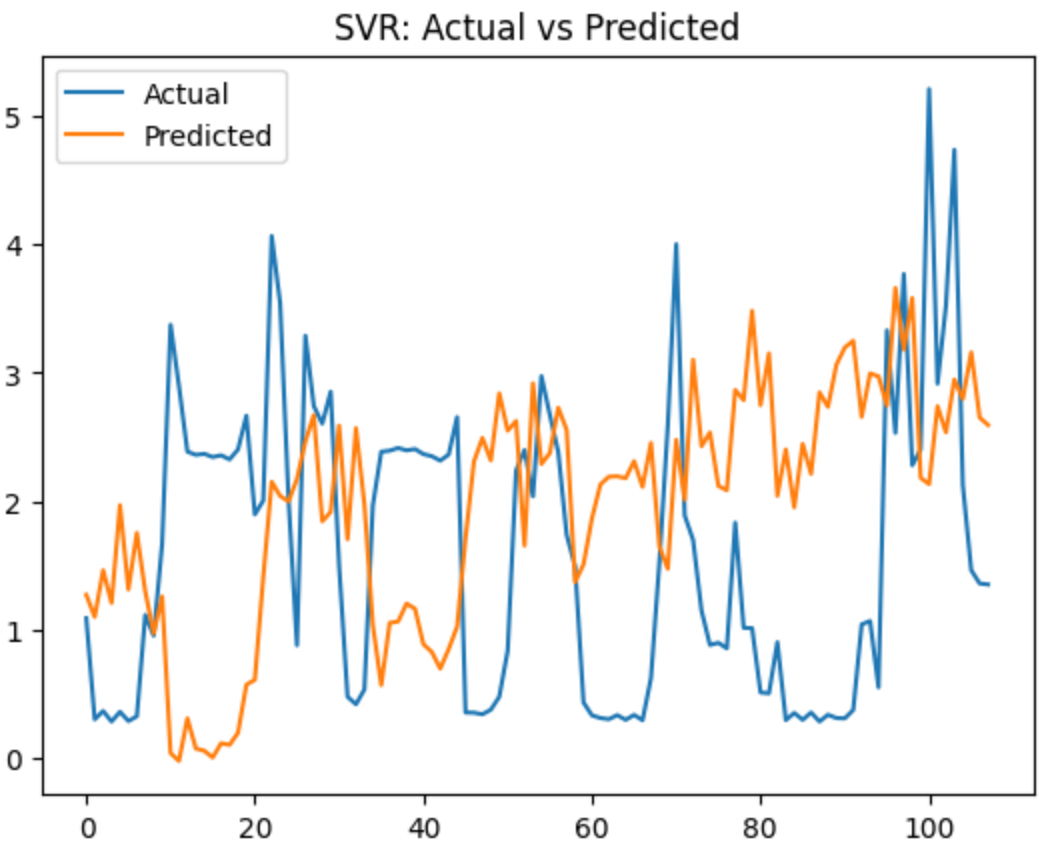
svr_model.fit(X_train, y_train)

y_pred_svr = svr_model.predict(X_test)

# Evaluation metrics
mae_svr = mean_absolute_error(y_test, y_pred_svr)
mse_svr = mean_squared_error(y_test, y_pred_svr)
rmse_svr = np.sqrt(mse_svr)
r2_svr = r2_score(y_test, y_pred_svr)

# ---- Graph: Actual vs Predicted ----
plt.figure()
plt.plot(y_test.values[:500], label="Actual")
plt.plot(y_pred_svr[:500], label="Predicted")
plt.title("SVR: Actual vs Predicted")
plt.legend()
plt.show()

print("SVR MAE:", round(mae_svr, 3))
print("SVR MSE:", round(mse_svr, 3))
print("SVR RMSE:", round(rmse_svr, 3))
print("SVR R²:", round(r2_svr, 3))
```



SVR MAE: 1.452

SVR MSE: 2.681

SVR RMSE: 1.637

SVR R^2: -1.03

Figure 7: Support Vector Regression (SVR)

(Source: Self-created in Google Colab)

The Support Vector Regression model is beneficial, as shown by the actual and predicted household energy consumption data, as illustrated in Figure 7. The SVR model has weaker predictive ability, with a high error and a negative R^2 value, indicating poor generalisation. This implies that SVR does not do well in capturing complicated temporal and nonlinear characteristics in the time series of energy consumption.

6. Model Selection and Design (LO1, LO2)

6.1 Train–Test Split Strategy

The standard chronological train test split was used to separate the data to maintain a chronological order. This method has ensured that the predictiveness of future observations is based on the basis of past information only. Time-series data cannot be split randomly, and in such a case, the information leakage will be present. According to Ali *et al.* (2024), a reliable forecasting evaluation requires sufficient temporal integrity of energy demand studies. This plan consequently helps legitimate performance evaluation in timely forecasting circumstances.

6.2 Selected Machine Learning Models

Three forecasting models have been chosen to represent different degrees of complexity. Linear Regression is a simple method of the model of a linear relationship. Random Forest Regressor takes non-linear interactions. Moreover, it was also thought to compare a conceptual deep learning model. According to Hussain *et al.* the use of both a simple and advanced model allows the thorough assessment of the forecasting strength of energy consumption data (2025).

6.3 Justification of Model Choices

All the models were chosen to strike a balance between interpretability and predictive capacity. Linear Regression provides flexibility and transparency. Random Forest has high resistance to non-linearity and noise. Long-term dependencies Long-term dependencies are modelled using conceptual deep learning models that incur more computational resources. According to Nguyen *et al.* (2025), comparative modelling improves the insights into the performance trade-offs in energy forecasting. Therefore, such decisions will aid the critical analysis that is consistent with the study aims.

7. Code Implementation (LO1)

7.1 Implementation Environment

Python implementation in the Google Colab environment was used to implement the models in a manner that could be scaled, and results could be reproduced. The manipulation and preprocessing of data were carried out with pandas, with visualisations created with matplotlib. The scikit-learn machine learning models were used to guarantee regular training and evaluation processes. According to Wilfling (2023), Python-based frameworks can be used to customise transparent feature development and a model of energy systems analysis development. Such an atmosphere encouraged effective experimentation and good records of forecasting procedures.

7.2 Model Implementation Workflow

The workflow used in implementation entailed structured data preparation, training of the model and the generation of predictions. The engineered and preprocessed data were divided into chronological sets, and then they were model-fitted. Both models were trained based on past observations and forecasted on unknown test data. In energy prediction studies, Abdullah *et al.* (2024) have mentioned that systematic training and validation processes are essential to have meaningful performance comparisons. This process guarantees methodological consistency and a valid assessment of the forecasting results.

8. Model Evaluation and Results (LO1, LO2)

8.1 Evaluation Metrics

Mean Absolute Error, Root Mean Squared Error, and R^2 were used to evaluate model performance. These measures have the effect of quantifying average prediction deviation; large errors are penalised, and explanatory power is respectively. According to Kim *et al.* (2023), the combination of error-based and variance-based metrics offers in-depth information regarding the accuracy of forecasting. Thus, the combination of the measures allowed for such strong measures of the performance of the predictability of the domestic energy consumption forecast.

8.2 Model Performance Comparison

Comparison of performance showed that the random forest model always outperformed the Linear Regression in all the evaluation criteria, which implies that it has a better management of non-linear consumption patterns. Linear Regression showed greater error values, which is indicative of weak modelling flexibility. According to Ghazal (2022), ensemble-based models are usually more accurate in challenging energy forecasting problems. Therefore, mathematical findings took a leading role in demonstrating the effectiveness of non-linear modelling in the capacity to predict household electricity demand.

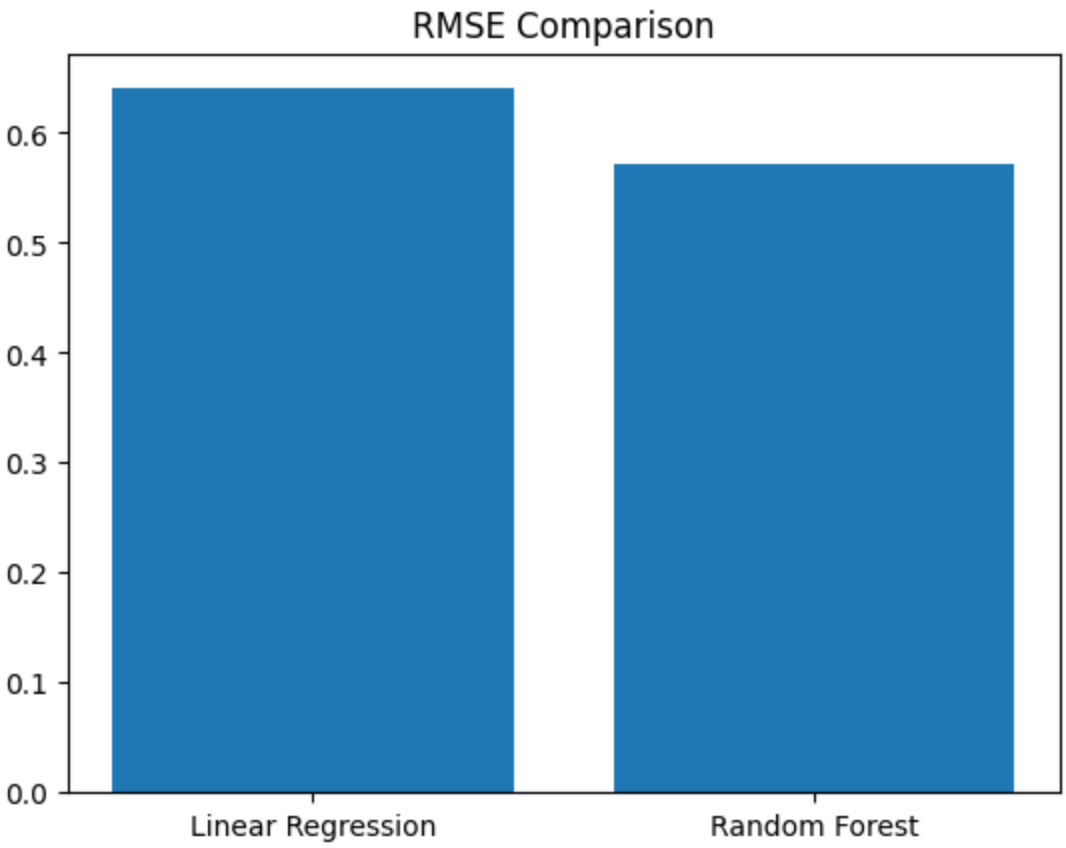


Figure 8: RMSE Comparison

(Source: Self-created in Google Colab)

This is a comparison of the losses of RMSE models. Random Forest has a lower RMSE (0.572) than Linear Regression (0.640), which means that it has a better predictive power. This comparison is clear about the superiority of ensemble learning methods compared to linear models on more complicated household energy time-series data.

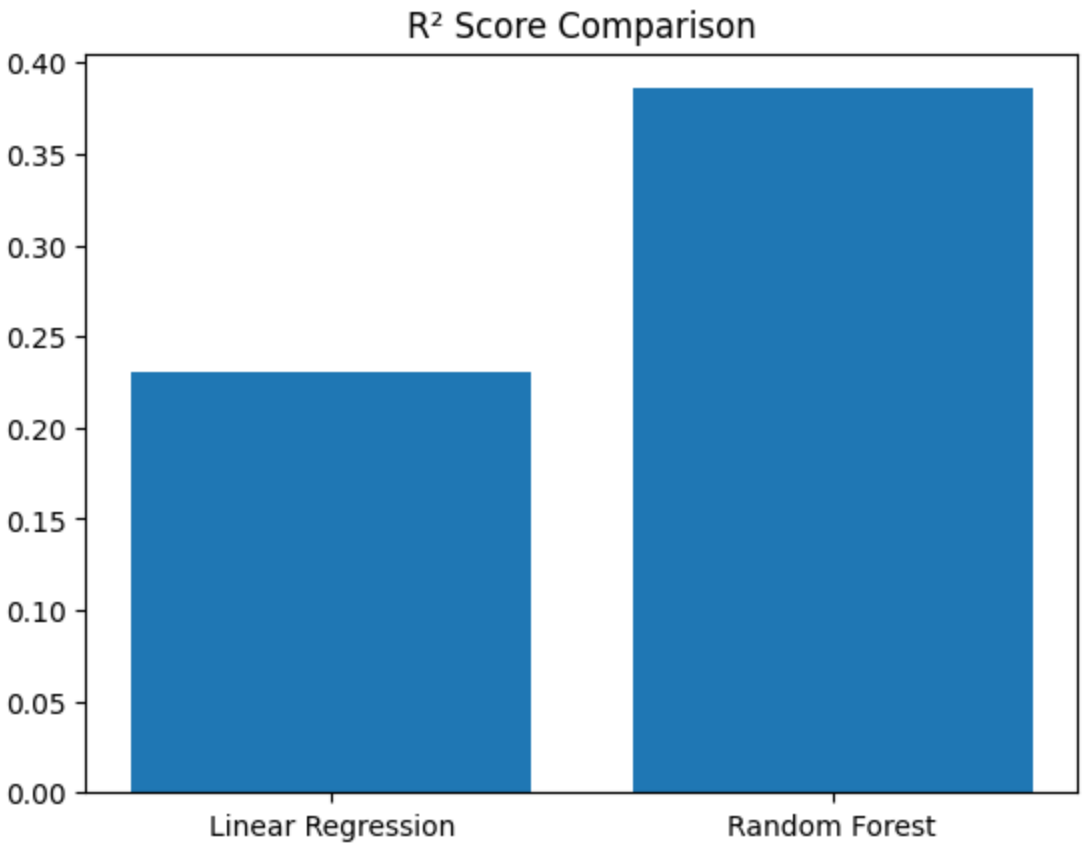


Figure 9: R^2 Score Comparison

(Source: Self-created in Google Colab)

Figure 8 gives the R^2 score of both models. Random Forest has a higher R^2 (0.386) than Linear Regression (0.230), which implies more variance. This validates the use of Random Forest as the most effective model in general, with better explanatory and predictive variables in predicting house energy consumption.

8.3 Visual Evaluation

Actual-versus-predicted plots indicated a visual analysis that had greater alignment of the observed and predicted values in the case of the Random Forest model. The garbage distributions also provided an indication of less variance and fewer severe errors. According to Olu-Ajayi *et al.* (2022), graphical diagnostics are useful in several complementary numerical metrics since these tools help to demonstrate bias-bearing systematic prediction. These graphical results contributed to the numerical results of better model accuracy and stability.

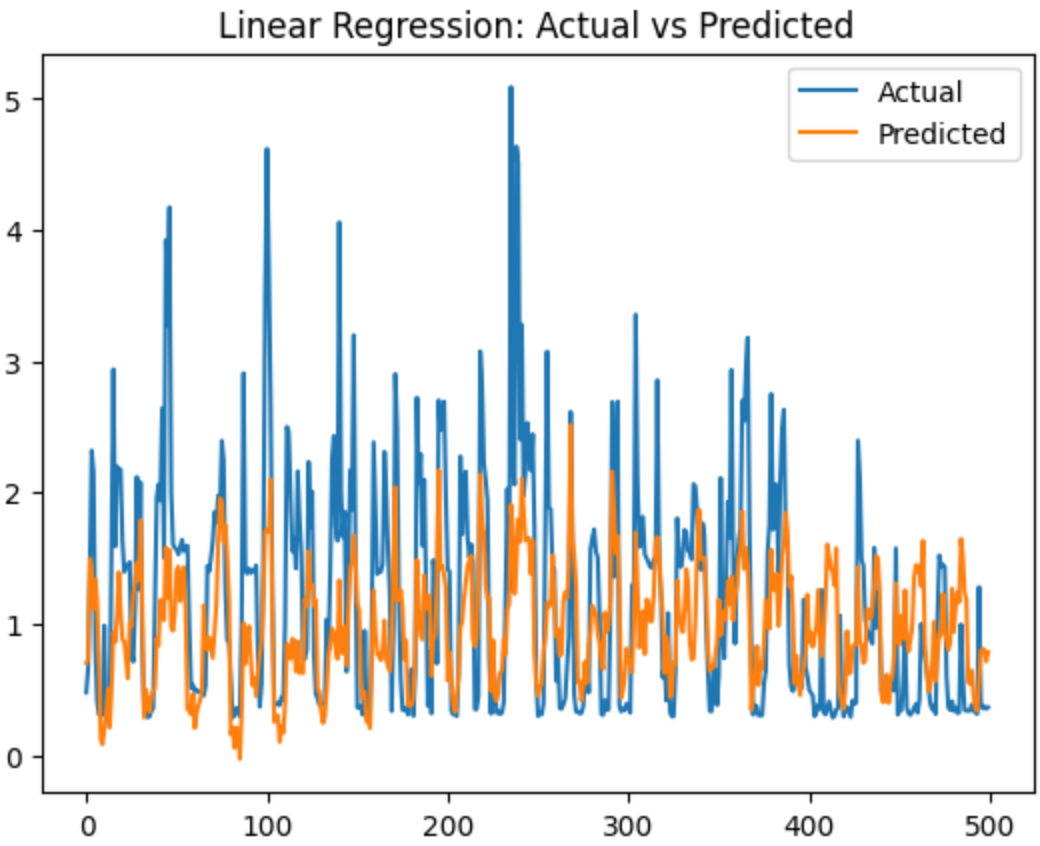


Figure 10: Linear Regression: Actual vs Predicted

(Source: Self-created in Google Colab)

This value is the comparison of the actual and the predicted power consumption based on Linear Regression. The model has an MAE of 0.52, RMSE of 0.64, and R 2 =0.23, which means that it has little explanatory power. The loss of orthogonality is the failure of the model to account for the non-linear patterns of consumption.

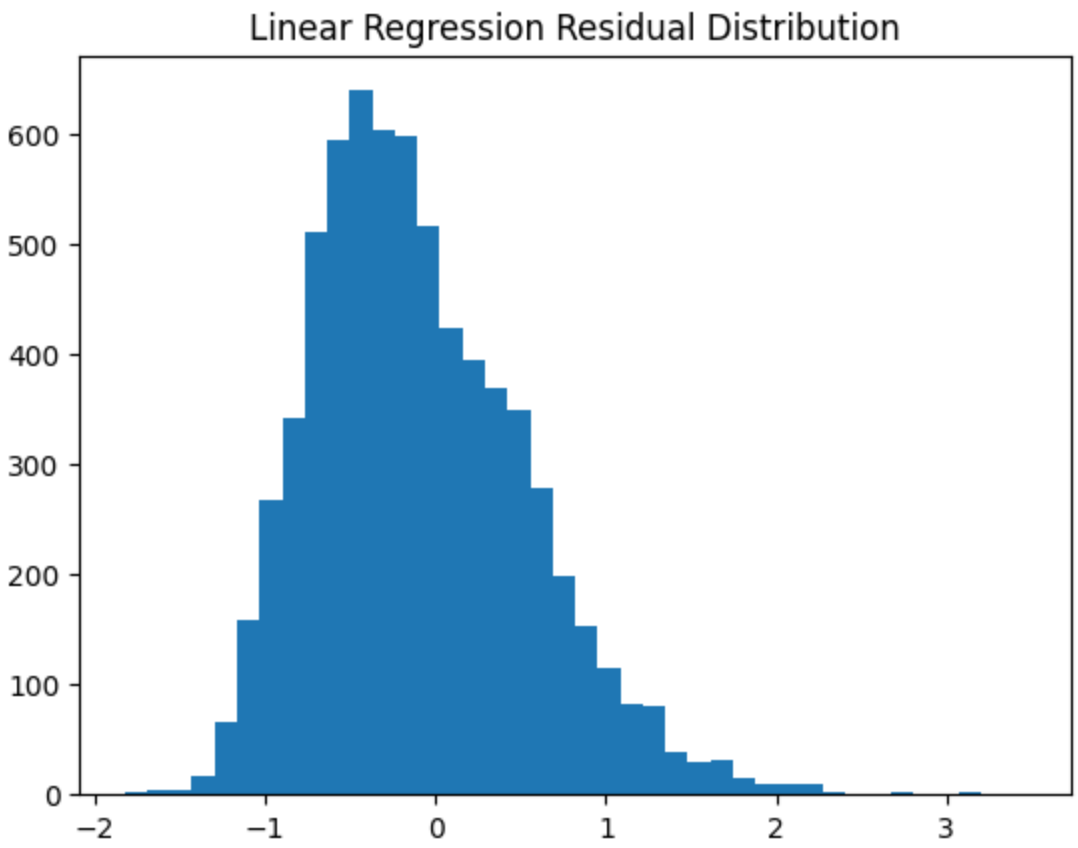


Figure 11: Linear Regression Residual Distribution

(Source: Self-created in Google Colab)

Figure 10 represents the residuals of the Linear Regression model distribution. The value of No (-0.0941) indicates low-level systematic under-prediction. The proliferation of residuals points towards patients of the model and solidifies the restriction of the linear assumption of the model to complex household energy consumption behaviour.

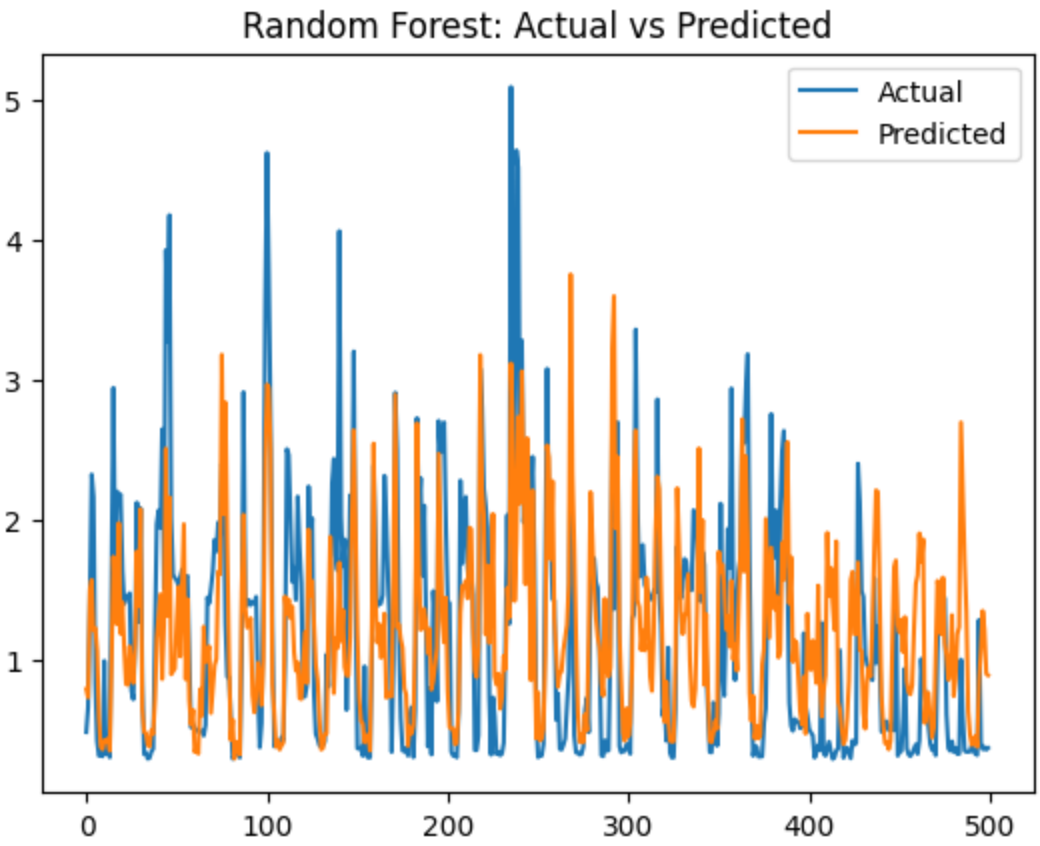


Figure 12: Random Forest: Actual vs Predicted

(Source: Self-created in Google Colab)

This value depicts the prediction of Random Forests against the reality of consumption. The model is more effective, with an MAE of 0.412, RMSE of 0.572, as well as R 2 of 0.386. The fact that there is closer agreement between the predicted and actual values shows that there is better coverage of the non-linear relationship in the data.

8.4 Best-Performing Model

Random Forest Regressor turned out to be the most successful model in terms of error and explanatory power. Its ensemble arrangement allowed learning complicated interactions between temporal and electrical characteristics effectively. According to Kamoona *et al.* (2023), ensemble learning schemes contribute to the generalisation in energy demand forecasting applications. Thus, this model was found to be the most appropriate one in predicting the consumption of household energy in the context of this paper.

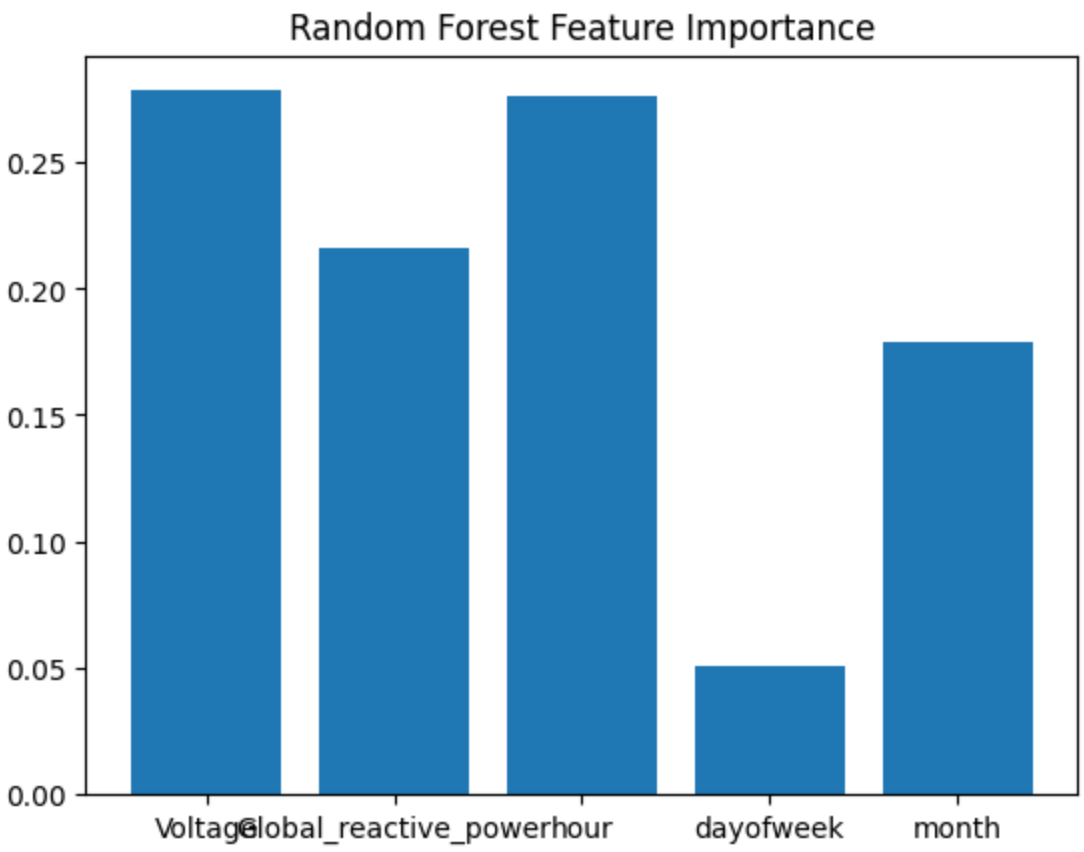


Figure 13: Random Forest Feature Importance

(Source: Self-created in Google Colab)

Figure 12 is a ranking of predictors by their contribution to Random Forest performance. The strongest influence is on voltage (0.278) and hour (0.276), then Global reactive power (0.216) and month (0.179). This verifies the significance of electrical features and temporal features in energy forecasting.

[6] ✓ 0s

```
# MSE for Linear Regression


mse_lr = mean_squared_error(y_test, y_pred_lr)
print("Linear Regression MSE:", round(mse_lr, 3))
```

```
Linear Regression MSE: 0.715
```

[7] ✓ 0s

```
# MSE for Random Forest

mse_rf = mean_squared_error(y_test, y_pred_rf)
print("Random Forest MSE:", round(mse_rf, 3))
```

```
Random Forest MSE: 1.075
```

[8] ✓ 0s

```
# UPDATED MODEL COMPARISON TABLE


results = pd.DataFrame({
    "Model": ["Linear Regression", "Random Forest", "SVR"],
    "MAE": [mae_lr, mae_rf, mae_svr],
    "MSE": [mse_lr, mse_rf, mse_svr],
    "RMSE": [rmse_lr, rmse_rf, rmse_svr],
    "R2": [r2_lr, r2_rf, r2_svr]
})

display(results)
```

	Model	MAE	MSE	RMSE	R2
0	Linear Regression	0.719071	0.715366	0.845793	0.458237
1	Random Forest	0.909855	1.074551	1.036605	0.186218
2	SVR	1.451909	2.680615	1.637258	-1.030091

Figure 14: MSE for Linear Regression, Random Forest, MODEL COMPARISON TABLE

Figure 14 shows the results of the Mean Squared error and a general comparison of the three models. Linear Regression has the lowest MSE thus showing the best predictive ability, then

random forest. When used with the highest error, SVR is not the most suitable due to the fact that it is not suitable for this forecasting of household energy consumption activity.

9. Conclusion and Future Work (LO2)

9.1 Summary of Findings

As revealed in the analysis, machine learning models were good at modelling temporal variation in household electricity. Random Forest performed better than linear baselines, leading to less forecast error and enhancing explanatory power. According to Kim *et al.* (2023), ensemble learning helps to increase the degree of strength in demand forecasting. It is generally found that it is eligible to predict.

9.2 Practical Implications

These findings show practical usefulness in energy management, where they are used to assist in demand planning, load balancing and efficiency in operations of smart grids. Better short-term predictions give the ability to make proactive decisions and optimise resources. According to Alzoubi (2022), intelligent energy management systems are empowered by proper prediction of households on a household level. According to Habib and Hossain (2024), feature spaces that have been enriched are highly predictive. These would increase scalability and applicability selectively. Accordingly, the forecasting provided by machine learning can provide useful operational insights.

9.3 Limitations

The study is limited in a number of ways. The sample only covers one household, and it is not likely to generalise to other regions and consumption patterns. Also, it is possible that engineered features can fail to capture external influences. According to Wen and Liu (2025), unobserved variables of the context limit the predictive accuracy. Other restrictions to adaptability in changing conditions include model assumptions and fixed hyperparameters.

9.4 Future Improvements

In the work of the future, the forecasting strength should be improved with the use of weather variables and occupancy indicators. More complex deep learning models can be used to better model long-term dependencies, and multivariate forecasting can be used to model cross-variable interactions.

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Appendices

Appendix 1:

Google Colab Link

https://drive.google.com/file/d/1KM7wLrdyKPBtQuf37nYdbB5j-sDP6MZT/view?usp=sharing

Github Link

https://github.com/minish-sarav/machinelearning-assignment

Appendix 2: Python Full Code

```
# Time-Series Forecasting for Household Energy Consumption
# Machine Learning Fundamentals – Practical Skills Assessment



# 1. IMPORT LIBRARIES

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor

import warnings
warnings.filterwarnings("ignore")



            # 2. LOAD DATA

            file_path = "/content/household_power_consumption.txt"

            df = pd.read_csv(
                file_path,
                sep=";",
                na_values="?",
                low_memory=False
            )
```

```
# 3. DATA PREPROCESSING


# Convert Date & Time to Datetime
df["Datetime"] = pd.to_datetime(
    df["Date"] + " " + df["Time"],
    format="%d/%m/%Y %H:%M:%S"
)

df.drop(["Date", "Time"], axis=1, inplace=True)
df.set_index("Datetime", inplace=True)

# Convert all columns to numeric
df = df.apply(pd.to_numeric, errors="coerce")

print("\nMissing values BEFORE cleaning:")
print(df.isnull().sum())

# Drop original missing rows
df = df.dropna()

print("\nDataset shape after initial cleaning:", df.shape)


# 4. RESAMPLING (HOURLY)

df_hourly = df.resample("h").mean()

# Handle NaNs introduced by resampling
df_hourly = df_hourly.ffill().bfill()

# Final safety check
df_hourly = df_hourly.dropna()

print("\nHourly dataset shape after NaN handling:", df_hourly.shape)
print("Any NaNs left?", df_hourly.isnull().sum().sum())
```

```
# SECTION 1: EXPLORATORY DATA ANALYSIS (EDA)


# ---- Graph 1: Global Active Power over time ----
plt.figure()
plt.plot(df_hourly.index, df_hourly["Global_active_power"])
mean_gap = df_hourly["Global_active_power"].mean()
plt.axhline(mean_gap)
plt.title("Hourly Global Active Power Consumption")
plt.xlabel("Time")
plt.ylabel("kW")
plt.show()

print("Mean Global Active Power (kW):", round(mean_gap, 3))

# ---- Graph 2: Voltage over time ----
plt.figure()
plt.plot(df_hourly.index, df_hourly["Voltage"])
mean_voltage = df_hourly["Voltage"].mean()
plt.axhline(mean_voltage)
plt.title("Hourly Voltage Trend")
plt.xlabel("Time")
plt.ylabel("Voltage (V)")
plt.show()

print("Mean Voltage (V):", round(mean_voltage, 2))
```

```python
# SECTION 2: SEASONALITY & DISTRIBUTION


# ---- Graph 1: Hourly average consumption ----
hourly_avg = df_hourly.groupby(df_hourly.index.hour)["Global_active_power"].mean()

plt.figure()
plt.plot(hourly_avg.index, hourly_avg.values)
plt.title("Average Power Consumption by Hour of Day")
plt.xlabel("Hour")
plt.ylabel("kW")
plt.show()

print("Peak Hour Consumption (kW):", round(hourly_avg.max(), 3))

# ---- Graph 2: Distribution ----
plt.figure()
plt.hist(df_hourly["Global_active_power"], bins=40)
plt.title("Distribution of Global Active Power")
plt.xlabel("kW")
plt.ylabel("Frequency")
plt.show()

print("Median Power Consumption (kW):", round(df_hourly["Global_active_power"].median(), 3))
```

```
# SECTION 3: FEATURE ENGINEERING


df_feat = df_hourly.copy()

df_feat["hour"] = df_feat.index.hour
df_feat["dayofweek"] = df_feat.index.dayofweek
df_feat["month"] = df_feat.index.month

# ---- Graph 1: Day-of-week trend ----
weekday_avg = df_feat.groupby("dayofweek")["Global_active_power"].mean()

plt.figure()
plt.plot(weekday_avg.index, weekday_avg.values)
plt.title("Average Consumption by Day of Week (0=Mon)")
plt.xlabel("Day")
plt.ylabel("kW")
plt.show()

print("Highest weekday consumption (kW):", round(weekday_avg.max(), 3))

# ---- Graph 2: Monthly trend ----
monthly_avg = df_feat.groupby("month")["Global_active_power"].mean()

plt.figure()
plt.plot(monthly_avg.index, monthly_avg.values)
plt.title("Average Monthly Power Consumption")
plt.xlabel("Month")
plt.ylabel("kW")
plt.show()

print("Highest monthly consumption (kW):", round(monthly_avg.max(), 3))
```

```
# SECTION 4: TRAIN-TEST SPLIT


target = "Global_active_power"
features = ["Voltage", "Global_reactive_power", "hour", "dayofweek", "month"]

X = df_feat[features]
y = df_feat[target]

# Final NaN safety check (critical)
assert X.isnull().sum().sum() == 0, "NaNs found in features!"
assert y.isnull().sum() == 0, "NaNs found in target!"

split_index = int(len(df_feat) * 0.8)

X_train, X_test = X.iloc[:split_index], X.iloc[split_index:]
y_train, y_test = y.iloc[:split_index], y.iloc[split_index:]

print("Training samples:", X_train.shape)
print("Testing samples:", X_test.shape)
```

```
# SECTION 5: MODEL 1 – LINEAR REGRESSION


lr = LinearRegression()
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

mae_lr = mean_absolute_error(y_test, y_pred_lr)
rmse_lr = np.sqrt(mean_squared_error(y_test, y_pred_lr))
r2_lr = r2_score(y_test, y_pred_lr)

# ---- Graph 1: Actual vs Predicted ----
plt.figure()
plt.plot(y_test.values[:500], label="Actual")
plt.plot(y_pred_lr[:500], label="Predicted")
plt.title("Linear Regression: Actual vs Predicted")
plt.legend()
plt.show()

print("Linear Regression MAE:", round(mae_lr, 3))
print("Linear Regression RMSE:", round(rmse_lr, 3))
print("Linear Regression R²:", round(r2_lr, 3))


# ---- Graph 2: Residuals ----
residuals_lr = y_test - y_pred_lr

plt.figure()
plt.hist(residuals_lr, bins=40)
plt.title("Linear Regression Residual Distribution")
plt.show()

print("Mean Residual (LR):", round(residuals_lr.mean(), 4))
```

```
# SECTION 6: MODEL 2 – RANDOM FOREST


rf = RandomForestRegressor(
    n_estimators=100,
    max_depth=15,
    random_state=42,
    n_jobs=-1
)

rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)

mae_rf = mean_absolute_error(y_test, y_pred_rf)
rmse_rf = np.sqrt(mean_squared_error(y_test, y_pred_rf))
r2_rf = r2_score(y_test, y_pred_rf)

# ---- Graph 1: Actual vs Predicted ----
plt.figure()
plt.plot(y_test.values[:500], label="Actual")
plt.plot(y_pred_rf[:500], label="Predicted")
plt.title("Random Forest: Actual vs Predicted")
plt.legend()
plt.show()

print("Random Forest MAE:", round(mae_rf, 3))
print("Random Forest RMSE:", round(rmse_rf, 3))
print("Random Forest R²:", round(r2_rf, 3))
```

```
# ---- Graph 2: Feature Importance ----
importances = rf.feature_importances_

plt.figure()
plt.bar(features, importances)
plt.title("Random Forest Feature Importance")
plt.show()

for f, imp in zip(features, importances):
    print(f"{f}: {round(imp, 3)}")


# SECTION 7: MODEL COMPARISON


results = pd.DataFrame({
    "Model": ["Linear Regression", "Random Forest"],
    "MAE": [mae_lr, mae_rf],
    "RMSE": [rmse_lr, rmse_rf],
    "R2": [r2_lr, r2_rf]
})

display(results)
```

```
# ---- Graph 1: RMSE comparison ----
plt.figure()
plt.bar(results["Model"], results["RMSE"])
plt.title("RMSE Comparison")
plt.show()

# ---- Graph 2: R² comparison ----
plt.figure()
plt.bar(results["Model"], results["R2"])
plt.title("R² Score Comparison")
plt.show()

print("Best Model based on RMSE:", results.loc[results["RMSE"].idxmin(), "Model"])
```